Supplementary Material

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This document supplements *"When Super-Resolution Hurts: A Controlled Study of Ear-Based Gender Classification under Deployment Constraints"* (DASA'26). It contains three tables: Table S1 extends Table IV of the main paper to all input variants, Table S2 reports the per-class metrics behind the recall and balanced-accuracy statements in Section IV-B, and Table S3 reports the label-free adaptation results behind Section IV-F.

S1. Protocol A vs. Protocol B, all model×variant combinations

Table IV of the main paper reports only the Original row of this comparison per backbone. Table S1 gives the complete table for all 6 backbones × 4 input variants (24 rows): Protocol A accuracy, Protocol B accuracy and macro-F1 (mean ± std over 5 folds), and the accuracy gap in percentage points. All rows use the same-checkpoint protocol: each backbone is trained once on Original images of the respective protocol under the locked configuration and tested on all four inputs. Protocol A holds out 10% of its training half for early stopping; its test half is the same as in the legacy split.

Table S1: Protocol A vs. Protocol B accuracy, same-checkpoint, all model×variant combinations.

Model	Variant	Proto. A (%)	Proto. B (%)	Proto. B macro-F1 (%)	Gap (pp)
VGG19	Original	93.16	90.74±0.87	90.09±1.07	2.42
VGG19	Bicubic	93.12	90.70±0.90	90.04±1.10	2.43
VGG19	Real-ESRGAN	89.53	88.78±0.68	88.20±0.62	0.74
VGG19	SwinIR	90.09	88.94±0.33	88.35±0.35	1.15
MobileNetV2	Original	93.90	91.76±1.32	91.17±1.42	2.14
MobileNetV2	Bicubic	93.60	91.71±1.21	91.10±1.31	1.89
MobileNetV2	Real-ESRGAN	89.64	88.16±1.38	87.70±1.35	1.48
MobileNetV2	SwinIR	90.03	88.83±0.79	88.37±0.78	1.20
ResNet50	Original	93.99	91.91±1.32	91.37±1.49	2.08
ResNet50	Bicubic	93.62	91.83±1.36	91.27±1.53	1.78
ResNet50	Real-ESRGAN	90.01	88.86±2.13	88.40±2.15	1.15
ResNet50	SwinIR	90.64	89.42±1.72	88.95±1.76	1.23
EfficientNet-B0	Original	94.84	92.61±0.79	92.12±0.92	2.23
EfficientNet-B0	Bicubic	94.76	92.58±0.81	92.08±0.96	2.18
EfficientNet-B0	Real-ESRGAN	90.27	86.17±4.48	85.83±4.32	4.10
EfficientNet-B0	SwinIR	90.41	85.79±4.58	85.47±4.42	4.62
Swin-T	Original	94.97	92.35±1.69	91.79±1.95	2.63
Swin-T	Bicubic	94.90	92.33±1.73	91.77±1.99	2.57
Swin-T	Real-ESRGAN	94.69	90.87±1.82	90.38±1.80	3.81
Swin-T	SwinIR	94.75	90.74±1.97	90.23±1.97	4.01

Table S1: (continued)

Model	Variant	Proto. A (%)	Proto. B (%)	Proto. B macro-F1 (%)	Gap (pp)
MaxViT-T	Original	95.73	92.72±1.50	92.24±1.62	3.01
MaxViT-T	Bicubic	95.67	92.77±1.38	92.29±1.48	2.90
MaxViT-T	Real-ESRGAN	93.83	89.87±2.23	89.48±2.15	3.95
MaxViT-T	SwinIR	93.62	90.44±1.89	90.02±1.82	3.18

Summary. The Protocol A→B gap is positive for all 24 combinations (range 0.74–4.62 pp, mean 2.45 pp). The mean gap is similar across input variants (Original 2.42, Bicubic 2.29, Real-ESRGAN 2.54, SwinIR 2.56 pp), but it varies more across backbones for the SR inputs (0.74–4.62 pp) than for Original and Bicubic (1.78–3.01 pp), because the same-checkpoint SR penalty differs between protocols for some backbones. Early-stopping Protocol A on its test half instead of the held-out split would have raised its Original accuracy by a further 0.5 pp on average. The inflation from image-level splitting is therefore not specific to one input variant. Absolute accuracies reported under an image-level split, as in the original EarVN1.0 gender study, should not be compared directly with subject-disjoint numbers, with or without super-resolution.

S2. Per-class metrics under the same-checkpoint protocol

Table S2 lists accuracy, macro-F1, balanced accuracy, and male and female recall for every backbone and input variant under Protocol B (mean ± std over the 5 subject-disjoint folds; each backbone uses one Original-trained checkpoint per fold). The last three rows give the mean change versus Original over the six backbones, which are the values quoted in Section IV-B of the main paper. Male recall falls and female recall rises for all six backbones under both Real-ESRGAN and SwinIR; under Bicubic, male recall falls for two backbones and female recall rises for one, with changes below 0.5 points.

Table S2: Per-class metrics (%) under the same-checkpoint protocol, Protocol B, mean ± std over 5 folds.

Model	Variant	Accuracy	Macro-F1	Bal. acc.	Recall (male)	Recall (female)
VGG19	Original	90.74±0.87	90.09±1.07	89.78±1.53	93.73±1.81	85.83±4.58
VGG19	Bicubic	90.70±0.90	90.04±1.10	89.75±1.57	93.67±1.77	85.82±4.62
VGG19	Real-ESRGAN	88.78±0.68	88.20±0.62	88.54±0.88	89.45±3.51	87.63±4.82
VGG19	SwinIR	88.94±0.33	88.35±0.35	88.69±1.02	89.63±3.57	87.75±5.51
MobileNetV2	Original	91.76±1.32	91.17±1.42	90.77±1.42	94.91±2.08	86.63±2.88
MobileNetV2	Bicubic	91.71±1.21	91.10±1.31	90.63±1.35	95.16±1.80	86.10±2.74
MobileNetV2	Real-ESRGAN	88.16±1.38	87.70±1.35	88.56±1.11	86.80±2.84	90.32±2.23
MobileNetV2	SwinIR	88.83±0.79	88.37±0.78	89.14±0.85	87.77±2.38	90.51±3.12
ResNet50	Original	91.91±1.32	91.37±1.49	91.17±1.74	94.25±1.54	88.09±3.96
ResNet50	Bicubic	91.83±1.36	91.27±1.53	91.00±1.75	94.44±1.47	87.56±3.78
ResNet50	Real-ESRGAN	88.86±2.13	88.40±2.15	89.15±1.79	87.91±3.51	90.38±1.82
ResNet50	SwinIR	89.42±1.72	88.95±1.76	89.55±1.53	88.97±2.98	90.13±2.44
EfficientNet-B0	Original	92.61±0.79	92.12±0.92	91.90±1.28	94.84±2.16	88.96±4.18
EfficientNet-B0	Bicubic	92.58±0.81	92.08±0.96	91.85±1.39	94.88±2.17	88.81±4.48
EfficientNet-B0	Real-ESRGAN	86.17±4.48	85.83±4.32	87.36±3.26	82.29±8.60	92.42±2.75
EfficientNet-B0	SwinIR	85.79±4.58	85.47±4.42	87.15±3.30	81.34±9.00	92.96±3.54
Swin-T	Original	92.35±1.69	91.79±1.95	91.52±2.45	94.89±2.70	88.16±6.43
Swin-T	Bicubic	92.33±1.73	91.77±1.99	91.49±2.49	94.94±2.73	88.03±6.51
Swin-T	Real-ESRGAN	90.87±1.82	90.38±1.80	90.65±1.70	91.47±5.19	89.82±6.12

Table S2: (continued)

Model	Variant	Accuracy	Macro-F1	Bal. acc.	Recall (male)	Recall (female)
Swin-T	SwinIR	90.74±1.97	90.23±1.97	90.51±1.90	91.36±5.33	89.66±6.40
MaxViT-T	Original	92.72±1.50	92.24±1.62	92.02±1.73	94.93±2.85	89.11±4.31
MaxViT-T	Bicubic	92.77±1.38	92.29±1.48	92.09±1.58	94.92±2.70	89.26±4.03
MaxViT-T	Real-ESRGAN	89.87±2.23	89.48±2.15	90.31±1.61	88.39±4.86	92.22±3.20
MaxViT-T	SwinIR	90.44±1.89	90.02±1.82	90.64±1.39	89.73±4.42	91.55±3.49
Mean Δ vs Original	Bicubic	−0.03	−0.04	−0.06	+0.08	−0.20
Mean Δ vs Original	Real-ESRGAN	−3.23	−3.13	−2.10	−6.88	+2.67
Mean Δ vs Original	SwinIR	−2.99	−2.90	−1.92	−6.46	+2.63

S3. Label-free adaptation of the same checkpoint

Table S3 lists the results of adapting each Original-trained Protocol B checkpoint to SR inputs without labels or retraining, as described in Section III-C of the main paper. The adaptation set is each fold's validation images, used without labels. *Threshold* shifts the female-minus-male logit margin so that the fraction of SR inputs predicted female equals the fraction the same checkpoint predicts on the Original versions of those images. *AdaBN* re-estimates the BatchNorm statistics on the SR images and applies only to the four backbones with BatchNorm, so VGG19 and Swin-T have no AdaBN rows. ΔAcc is the accuracy change versus the same checkpoint on Original inputs; all values are means over the 5 folds.

Threshold recalibration improves on no adaptation in 53 of 60 backbone-fold-SR cases and AdaBN in 38 of 40. The remaining loss stays significant: fold-level 95% confidence intervals of ΔAcc are $[-1.92, -0.95]$ (Real-ESRGAN) and $[-1.67, -0.89]$ (SwinIR) for threshold recalibration, and $[-1.61, -0.76]$ and $[-0.87, -0.65]$ for AdaBN. Averaged over both SR methods and the six backbones, male recall rises from 87.93% without adaptation to 93.31% with threshold recalibration (Original: 94.59%). The summary rows give the values quoted in Section IV-F.

Table S3: Label-free adaptation under the same-checkpoint protocol, Protocol B, mean over 5 folds (%).

Model	SR	Method	ΔAcc (pp)	Bal. acc.	Recall (male)	Recall (female)
VGG19	Real-ESRGAN	None	−1.96	88.54	89.45	87.63
VGG19	Real-ESRGAN	Threshold	−1.14	88.58	92.78	84.39
VGG19	SwinIR	None	−1.79	88.69	89.63	87.75
VGG19	SwinIR	Threshold	−1.01	88.68	93.01	84.36
MobileNetV2	Real-ESRGAN	None	−3.60	88.56	86.80	90.32
MobileNetV2	Real-ESRGAN	Threshold	−1.82	88.86	93.37	84.34
MobileNetV2	Real-ESRGAN	AdaBN	−0.95	89.71	94.29	85.13
MobileNetV2	Real-ESRGAN	AdaBN+Thr.	−1.05	89.69	93.92	85.46
MobileNetV2	SwinIR	None	−2.93	89.14	87.77	90.51
MobileNetV2	SwinIR	Threshold	−1.24	89.51	93.75	85.26
MobileNetV2	SwinIR	AdaBN	−0.42	90.39	94.36	86.41
MobileNetV2	SwinIR	AdaBN+Thr.	−0.54	90.27	94.23	86.30
ResNet50	Real-ESRGAN	None	−3.05	89.15	87.92	90.38
ResNet50	Real-ESRGAN	Threshold	−1.47	89.61	93.01	86.21
ResNet50	Real-ESRGAN	AdaBN	−1.11	89.90	93.60	86.20
ResNet50	Real-ESRGAN	AdaBN+Thr.	−1.23	89.92	93.06	86.77
ResNet50	SwinIR	None	−2.49	89.55	88.97	90.13
ResNet50	SwinIR	Threshold	−1.53	89.70	92.47	86.94
ResNet50	SwinIR	AdaBN	−0.77	90.31	93.72	86.90
ResNet50	SwinIR	AdaBN+Thr.	−0.95	90.32	92.93	87.72

Table S3: (continued)

Model	SR	Method	ΔAcc (pp)	Bal. acc.	Recall (male)	Recall (female)
EfficientNet-B0	Real-ESRGAN	None	−6.44	87.35	82.29	92.41
EfficientNet-B0	Real-ESRGAN	Threshold	−2.57	89.17	92.79	85.55
EfficientNet-B0	Real-ESRGAN	AdaBN	−1.54	90.15	93.96	86.33
EfficientNet-B0	Real-ESRGAN	AdaBN+Thr.	−1.58	90.27	93.41	87.14
EfficientNet-B0	SwinIR	None	−6.82	87.15	81.34	92.96
EfficientNet-B0	SwinIR	Threshold	−2.27	89.52	92.91	86.12
EfficientNet-B0	SwinIR	AdaBN	−0.97	90.89	93.99	87.78
EfficientNet-B0	SwinIR	AdaBN+Thr.	−1.01	90.87	93.85	87.90
Swin-T	Real-ESRGAN	None	−1.47	90.65	91.47	89.82
Swin-T	Real-ESRGAN	Threshold	−0.52	91.03	94.28	87.78
Swin-T	SwinIR	None	−1.61	90.51	91.36	89.66
Swin-T	SwinIR	Threshold	−0.58	90.95	94.30	87.59
MaxViT-T	Real-ESRGAN	None	−2.84	90.31	88.40	92.22
MaxViT-T	Real-ESRGAN	Threshold	−1.07	91.02	93.61	88.42
MaxViT-T	Real-ESRGAN	AdaBN	−1.13	91.15	92.94	89.37
MaxViT-T	Real-ESRGAN	AdaBN+Thr.	−0.91	91.23	93.63	88.82
MaxViT-T	SwinIR	None	−2.28	90.64	89.73	91.55
MaxViT-T	SwinIR	Threshold	−1.07	91.07	93.43	88.71
MaxViT-T	SwinIR	AdaBN	−0.89	91.28	93.54	89.02
MaxViT-T	SwinIR	AdaBN+Thr.	−0.72	91.37	93.95	88.79
Mean, 6 backbones	Real-ESRGAN	None	−3.23	89.09	87.72	90.47
Mean, 6 backbones	Real-ESRGAN	Threshold	−1.43	89.71	93.31	86.12
Mean, 6 backbones	SwinIR	None	−2.99	89.28	88.13	90.42
Mean, 6 backbones	SwinIR	Threshold	−1.28	89.91	93.31	86.50
Mean, 4 BN backbones	Real-ESRGAN	None	−3.98	88.84	86.35	91.34
Mean, 4 BN backbones	Real-ESRGAN	AdaBN	−1.18	90.23	93.70	86.76
Mean, 4 BN backbones	SwinIR	None	−3.63	89.12	86.95	91.29
Mean, 4 BN backbones	SwinIR	AdaBN	−0.76	90.72	93.90	87.53
Original (reference)	—	—	—	91.19	94.59	87.79